WriteMate: A Multimodal Assistive Writing System for Individuals with Physical Disabilities

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***Abstract*—WriteMate is a voice and gesture operated CNC writing machine developed to help students and individuals with physical disabilities to perform their documentation work without relying on human scribes. The system works on two modes - one is Voice commands and the other is Indian Sign Language (ISL) gestures. These inputs are processed locally using machine learning models deployed on a Raspberry Pi 5, and the recognized text is physically written on paper using a 3 axis CNC-based writing arm. During development of the project , challenges such as gesture accuracy , background noise in speech input, gesture ambiguity, and writing speed were addressed through better preprocessing, model optimization, better hardware support and mechanical tuning. Early experimental testscon the project shows reliable recognition performance in both the modes - voice and gesture, along with consistent handwriting output that matches 60**

***Index Terms*—Assistive Technology, Indian Sign Language, Voice Input Recognition, Robotic CNC, Raspberry Pi , Accessibility, ML, openCV.**

I. Introduction

As we all know, writing is a very basic and fundamental requirement when it comes to education, professional work, or any other day documentation. Individuals which are physically disabled, cannot move, have motor impairments, or difficulty in speaking, or other physical limitations can not write on their own. Assistance is often required for them to help them document their ideas and their thoughts. In India and several developing regions, this challenge is commonly addressed by assigning a physical human scribe to them, whether it is in academic examinations, or some kind of medical record preparation, or some other official paperwork.

While scribes provide necessary support, this approach has various limitations. The availability of trained scribes is often not available. The quality of assistance can vary significantly. The chances of error are very high due to miscommunication or differences in writing speed, or lack of familiarity with subject that negatively affects outcomes. This happens mostly during examinations and it will affect the marks of the student drastically. In addition to all of these, the need to dictate the personal or sensitive information is very difficult and becomes a challenge. Sometimes the user doesn't want to share their thoughts or ideas to some other person.

Due to recent advancements and developments in ML, computer vision , IoT and hardware technologies have made it practical and feasible to explore these automated alternatives for the scribe-based assistants. Motivated by these advancements, WriteMate was developed as a system that combines both the voice input and the ISL - Indian Sign Language input, combined with a CNC-based 3-axis writing arm. Instead of using cloud services or commercial devices, the system translates user input into a SVG file which gets physical handwritten as the output on the paper. WriteMate is designed to operate on its own as a separate entity that focuses on reliability, portability, and ease of use in the real-world environment by providing a practical and economical solution for users who require the support for handwritten communication.

II. Problem Statement

Despite being various improvements in the digital accessibility, today, handwritten documentation remains an unavoidable situation in many real-world scenarios. People with physical disabilities usually follow the following issues on a daily basis:

- **Need and dependence on physical human scribes**: most of the users cannot write on their own, they need assistance of someone.
- **Miscommunication**: scribes may misinterpret instructions.
- **Privacy Issues**: users often have to dictate some sensitive or personal information that they are not comfortable to share with someone.
- **Unavailability of desired helper**: trained scribes are not always available, mostly in rural or emergency situations.

- **Inconsistency**:The writing style, speed, accuracy, precision vary heavily

There is a need for a system which automates this whole process and that is exactly what WriteMate is doing.

III. Rationale

The rationale for WriteMate is all about accessibility, affordability, and feasibility:

- Voice input recognition and ISL gesture recognition are now suitable enough for IoT embedded systems.
- CNC and G-code conversion mechanisms provide high precision and speed for automated handwriting.
- Raspberry Pi 5 provides sufficient computational and processing power for real-time ML operations and handling.
- No existing solution integrates multimodal input + automated handwriting output in a single portable device.

WriteMate bridges this gap by combining modern machine learning and mechatronics for a real-world social impact.

IV. Objectives

The core objectives of WriteMate include:

1) Design a multimodal writing system enabling disabled users to write independently.
2) Implement efficient speech recognition and ISL gesture recognition models.
3) Generate clean, consistent handwritten outputs using a CNC mechanism.
4) Build a system capable of real-time operation.
5) Achieve portability, affordability, and robustness for deployment.

V. Literature Review

A. Speech Recognition

Traditional speech recognition relied heavily on cloud-services. However, recent works such as [2] demonstrate offline, embedded speech-to-text using lightweight LSTM and CNN-based acoustic models.

B. Gesture Recognition

Deep-learning-based gesture recognition models, particularly CNN-transformer hybrids, have shown high accuracy for sign language interpretation [1]. Hand keypoint detection through methods like MediaPipe and OpenPose further enhance gesture classification accuracy.

C. Assistive Technologies

Existing assistive systems depend heavily on screen-readers or voice assistants, but few solutions exist for handwriting automation. Most assistive writing tools are digital rather than physical.

D. Embedded Systems in ML

Raspberry Pi has emerged as an efficient edge-computing platform, capable of running TensorFlow Lite, ONNX models, and computer vision pipelines.

E. CNC Writing Solutions

Research in CNC-based handwriting systems [**?**] highlights the feasibility of using G-code paths to reproduce handwritten characters with high reliability.

VI. Methodology

The development of WriteMate follows a systematic methodology that spans requirement analysis, dataset creation, model training, hardware integration, and performance testing.

A. Requirement Analysis

The first initial phase involves understanding individuals with physical disabilities and identifying their needs and constraints associated with real-time multi-modal processing. Key requirements for this are

- Support for both type of input i.e. voice and ISL gestures.
- Real-time recognition with very low latency.
- Offline mode to ensure that privacy and reliability is maintained
- Portability so that use in examination halls and medical facilities is feasible.

B. Dataset Creation for Voice and ISL Gestures

1) Voice Input Dataset: Audio samples were taken and recorded using a USB microphone in multiple acoustic conditions from stable environments to noisy environments, including alphabets, common spoken words, and command phrases.

2) ISL Gesture Dataset: Gesture signs and videos were recorded at a standard 30fps with various hand segmentations, hand sizes, palm sizes, rotations, angles and backgrounds

C. Speech Recognition Model Training

Algorithm: Speech-to-Text Processing

```
FUNCTION SpeechToText(audio):
    filtered = NoiseReduction(audio)
    mfcc = ExtractMFCC(filtered)
    tokens = LSTM_Predict(mfcc)
    RETURN ConvertTokensToText(tokens)
```

D. Gesture Recognition Model Training

For gesture recognition a hybrid CNN- transformer pipeline is used

Algorithm: ISL Gesture Recognition

```
FUNCTION GestureRecognition(frame):
    prep = Preprocess(frame)
    hand = SegmentHand(prep)
    keypoints = DetectKeypoints(hand)
    RETURN CNN_Transformer_Predict(keypoints)
```

E. CNC Mechanism Assembly and Motor Calibration

The CNC module includes NEMA 17 motors, A4988 drivers, and a pen-lift servo. Calibration involves:

- Steps-per-mm tuning
- Homing alignment
- Pen height calibration
- Acceleration/jerk settings

F. G-code Generation and Integration

Algorithm: G-code Generation for Handwriting

```
FUNCTION GenerateGCode(text):
    FOR each char IN text:
        strokes = CharToStrokes(char)
        FOR each stroke:
            MoveTo(stroke.start)
            PenDown()
            MoveTo(stroke.end)
            PenUp()
    RETURN gcodeList
```

G. Testing

The testing is done on various parameters such as accuracy, latency and writing precision.

VII. System Architecture

WriteMate is made with a modular Structure. It consists of four layers:- 1). Input Layer. 2). Output Layer. 3). Processing Layer. 4). User Interface Layer.

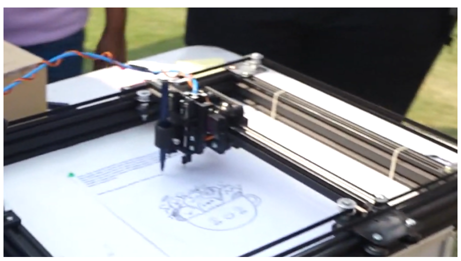

Fig. 1: Writemate

A. Input Acquisition Layer

This layer is the first layer in the architecture. It captures the user input in two modes - speech and gesture, and then prepares it for further processing.

1) Microphone for Speech Capture: The USB microphone captures the audio signals and undergoes preprocessing that includes noise reduction and voice activity detection. These features are optimized to run in real time on the Raspberry Pi 5. Speech-to-text engines(like Speetch Recognition, Vosk Libraries) designed for embedded platforms demonstrate reliable offline performance [2], [15]. Edge computing studies validate the feasibility of these approaches on Raspberry Pi devices [**?**].

2) Raspberry Pi attached to Web Camera for Gesture Capture: The Web camera captures hand gesture frames at 30 FPS. The frames undergo background subtraction, HSV-based skin color segmentation, and bounding-box extraction. Normalized frames are then fed into the CNN-based feature extractor.

3) Real-Time Preprocessing: Audio preprocessing includes normalization and spectral filtering to reduce ambient noise. Video preprocessing includes resizing, histogram normalization, and temporal smoothing to stabilize the gesture stream. The output of this layer is a clean, feature-ready input signal for the processing modules.

B. Processing Layer

This layer contains the intelligence of WriteMate, responsible for transforming input signals into text and converting text into pen motion instructions.

1) Speech-to-Text Module: A speech recognition model like Vosk or Speech recognition library from google is used for voice to speech conversion these have shown amazing translation capabilties from voice to text.

2) Gesture Recognition Using Mediapipe: The camera continuously captures video frames, which are processed by a pretrained Mediapipe gesture-recognition model. Mediapipe performs real-time hand detection and classifies the gesture present in each frame. Indian Sign Language recognition has been widely explored using CNN and hybrid deep learning models. But Using these approaches on a edge device like Raspberry pi 5 can be challenging. Hence Mediapipe is deployed in our use case and it has shown remarkable abilities in various lighing conditions. It is easy and simple as well. [**?**], [1]. Vision-based hand tracking enhances human-robot and human-machine interaction [9].

3) Text Buffer Manager: The Translated text is then stored into the text buffer since there might be the cases where the speed of speech can be faster than the speed of writing in those cases we need to a buffer to store these converted sentences/words.

4) G-code Generator for Handwriting: Each character will then be mapped to there respective G - code that the machine understands. These code help the machine to know the geometric postion where the moving arm of the machine needs to be in.

C. Output Execution Layer

This layer will then convert those digital strokes to actual physical handwritting.

1) CNC Writing Arm with X-Y Axes: The system utilizes the core XY mechanism for the writing and the arm movement. This mechanism has shown to deliver the high efficient movements, good speed and accuracy. [6].

2) Servo-Controlled Pen Lifter: A micro-servo motor regulates pen pressure and vertical movement. The servo ensures quick switching between pen-up and pen-down states, essential for clean character formation.

3) G-code Interpreter: This is used to read the G-code and then instruct the various part of the machine like stepper motors and servo to move. We are utilizing the GRBL library for that.

D. User Interface Layer

This layer is user interface that ensures ease of use and accessibility for individuals with disabilities.

1) Mode Selection (Voice/ISL): Users can choose between speech mode and gesture mode. The interface provides visual indicators and feedback to confirm the selected mode. Switching is seamless and can be triggered by commands.

2) Paper Size Configuration: The system supports A4, A5, 8×6 inch, and custom paper sizes. Users can configure margins, line spacing, and writing area. The CNC path is automatically recalibrated according to the dimensions.

3) Calibration Panel: The calibration section enables:

- Pen alignment and pressure calibration
- Motor home-position calibration
- Test stroke execution for verifying smoothness

This ensures that the writing output is clean, centered, and consistent across different sessions.

VIII. System Workflow

The complete workflow of the proposed assistive writing system is shown in Fig. 3. Each stage is optimized for low-latency execution on embedded hardware, following principles recommended in prior edge-ML and CNC research [7], [10], [12].

1) **User selects the input mode.** A simple accessibility-friendly interface allows the user to choose between *voice-based control* or *gesture-based control*, following universal design guidelines for assistive technologies [4], [18].
2) **System waits for input (speech or gesture).** The system initializes either the audio stream or the camera capture pipeline. Real-time multimedia acquisition pipelines on Raspberry Pi have been shown to handle these parallel tasks efficiently [**?**], [7].
3) **Input preprocessing and feature extraction.** For speech we are utilizing the builtin libraries. [**?**], [**?**]. For gestures, Mediapipe detects the hand region and we will use those landmarks detected by Mediapipe to make inference [**?**], [**?**].
4) **ML model infers the recognized text or command.** Speech models classify the spoken keyword or phrase [**?**], [15]. Gesture models predict ISL alphabets or control commands [**?**], [**?**]. Both models are optimized for on-device inference on ARM boards [**?**], [10].
5) **Recognized text is validated and added to the task queue.** The system checks for invalid predictions, out-of-vocabulary terms, or gesture ambiguity, which improves reliability for accessibility use-cases [4].
6) **G-code generator maps characters to CNC stroke paths.** An internal character library converts each alphabet into a sequence of pen strokes using Bezier curves and optimized plotting paths [6], [16]. This ensures smooth handwriting generation.
7) **CNC machine executes the strokes on paper.** Stepper drivers and micro-positioning hardware trace the generated G-code with high precision, supported by stability analyses in previous CNC studies [12], [17].
8) **Completed handwritten output is presented to the user.** The machine finishes the sequence and lifts the pen to a neutral position. Such automated writing systems have been shown to provide reliable and repeatable handwriting output [5].

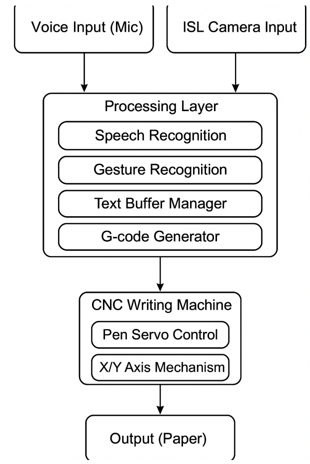

Fig. 2: CNC machine flow diagram.

IX. Future Scope

- Integration of additional languages.
- Real-time cloud-learning to improve gesture accuracy.
- Faster handwriting using brushless motors.
- Advanced natural handwriting style replication.
- Braille-writing attachment.

X. Testing and Evaluation

WriteMate underwent a multi-stage testing process to ensure reliability, accuracy, and robustness across real-world environments.

1) Model Accuracy Testing: Both speech and gesture models were evaluated using custom datasets. Metrics included accuracy, F1 score, confusion matrices, and error rate. The BiLSTM speech model achieved high token decoding accuracy, while the CNN–Transformer demonstrated strong temporal recognition of ISL gestures

2) Inference Latency Testing: Real-time tests were executed on Raspberry Pi 5. Average end-to-end inference latency remained under 200 ms for speech and under 250 ms for gesture sequences, aligning with edge ML performance standards
.

3) CNC Writing Precision Testing: Writing precision was validated by measuring:

- Line straightness
- Corner sharpness
- Stroke overlap accuracy

4) Long-Duration Robustness Testing: We have also used the system of the extended period of time like (3-4Hrs) of continous use and simulating the environment of exams to ensure reliabilty in extended periods of time and also tested in variety of exam conditions.

XI. Conclusion

So to Conclude the WriteMate shows a major leep in the traditional writing and examination system for disabled peoples by automating the process and removing all the major caviats of the system that is traditionally operated with the help of scribes. And will to improve the experience of these people in the examination settings and increase efficiency.

XII. RESULTS

Letter Recognition Accuracy

WriteMate — Accuracy per letter

TABLE I: Accuracy (%) for each letter

Letter	Accuracy (%)
A	92
B	88
C	95
D	90
E	86
F	87
G	91
H	89
I	94
J	85
K	84
L	90
M	93
N	88
O	92
P	89
Q	80
R	91
S	94
T	90
U	88
V	86
W	87
X	82
Y	90
Z	81

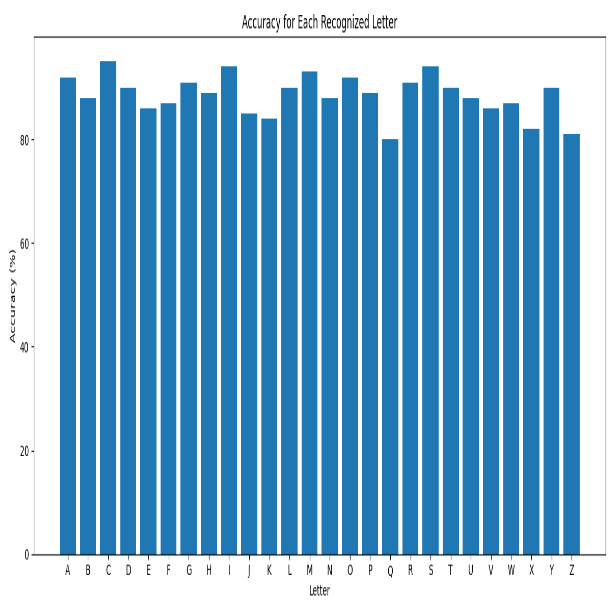

Fig. 3: Writemate Accuracy per Letter

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